



Wastewater Treatment Project

Introduction

Wastewater treatment plants play a crucial role in protecting the environment by removing contaminants from water before it is released back into nature. These systems help reduce pollution, protect ecosystems, and ensure that water resources remain safe for human use.

In this project, we use data-driven supervised learning methods to model wastewater treatment systems. By analyzing the relationship between input variables and output variables, we compare different models and determine which one provides the most accurate predictions.



Introduction

Hypothesis:

If the appropriate model is used and the data is properly standardized, then the RMSE (Root Mean Square Error) will be minimized.



Technical Review (Background)

Wastewater treatment involves major stages with different processes:

Stage 1 - Removal of large debris and heavy solids from incoming wastewater

Stage 2 - Removal of solid objects and particles through screening and settling tanks

Stage 3 - Use of bacteria and microorganisms to break down finer organic matter in wastewater

Stage 4 - Further physical and chemical filtration of wastewater using membranes and/or nanofilters + disinfection to kill pathogens in wastewater



Parameters

- The two types of parameters are system and modeling
- Input parameters are the flow rates (Q_{inf} , $Q_{air\ 1-5}$) and environmental factors such as temperature in the aerobic zone
- Output parameters are the dissolved oxygen (DO), and chemical levels of NO_3 and NH_4
- Modeling parameters are the parameters we changed to improve the model
 - Data split ratio
 - The error metric (RMSE)
 - Normalization
 - Framework

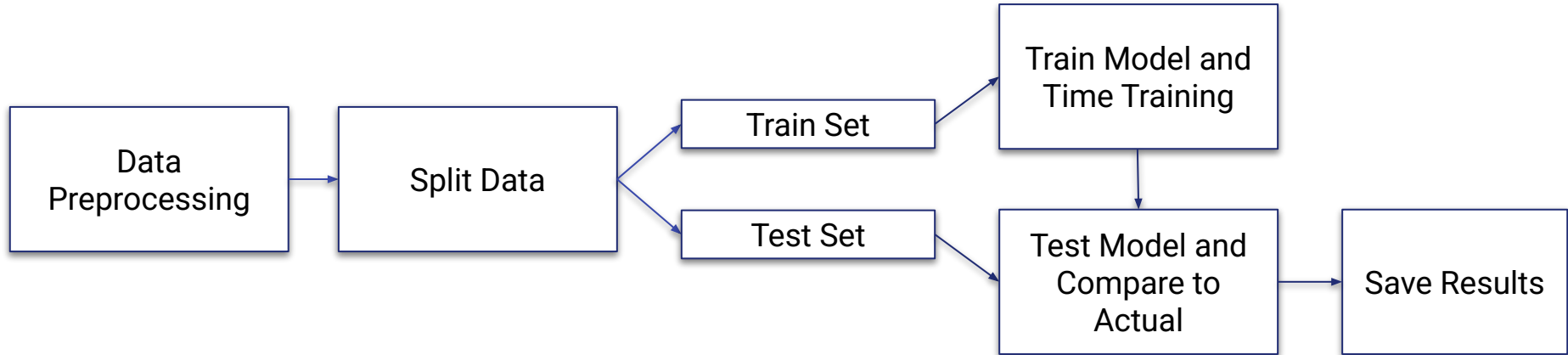


Outputs/Experimental Design

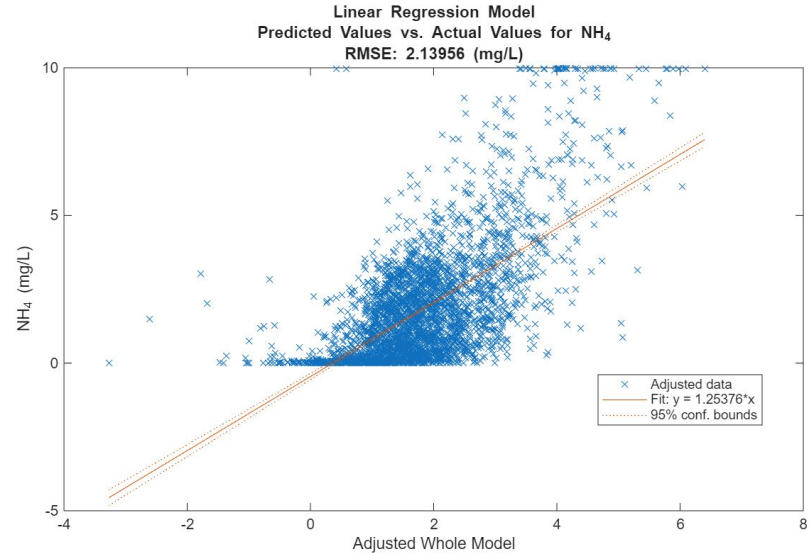
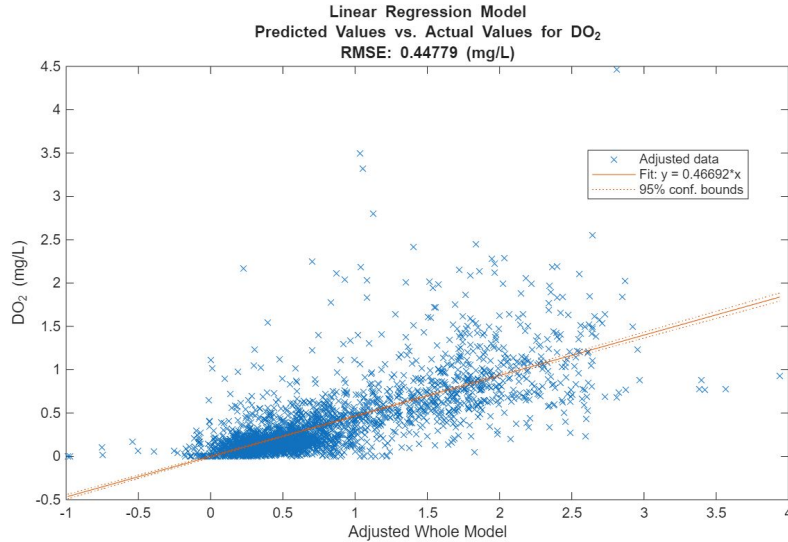
- Goal: predict 5 key water quality metrics of safety
- Dependent Variables
 - Dissolved oxygen (DO_{1-3})
 - Nitrate (NO_3)
 - Ammonium (NH_4)
- Success of the measurements is dependent on the RMSE value.



Model Results/MATLAB Programming - Flow Chart

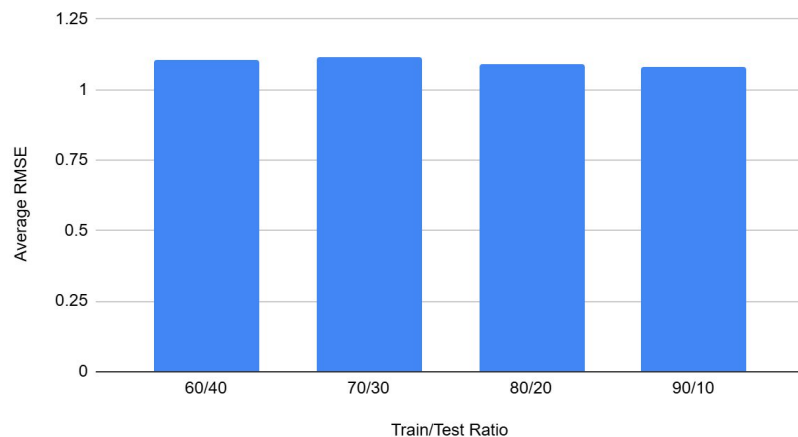


Model Results/MATLAB Programming - Visualization

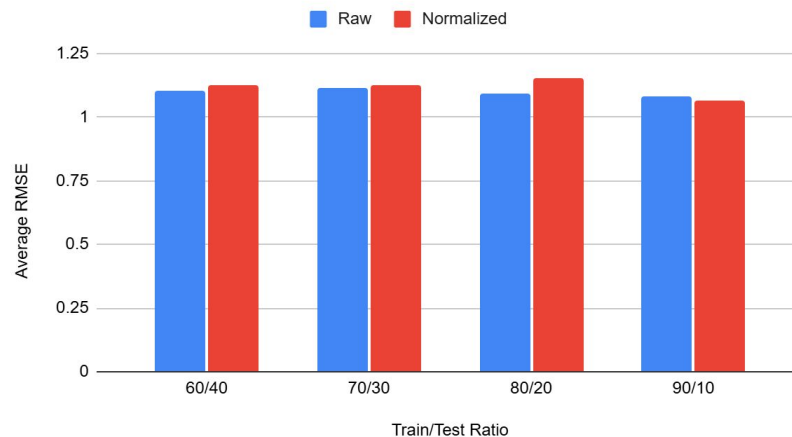


Model Results/MATLAB Programming - Follow Up Results

Average RMSE for Linear Regression Models of Various Train/Test Ratios



Average RMSE for Linear Regression Models of Various Train/Test Ratios



Analysis of Results

■ LR avg 0.838

■ GLR avg 0.743

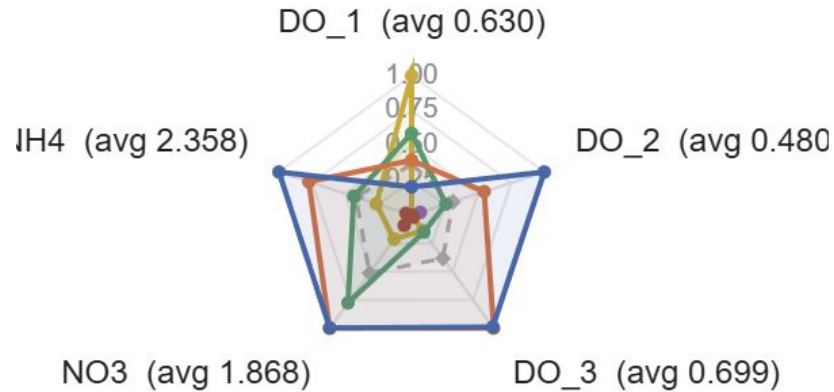
■ GPR avg 0.441

■ DT avg 0.030

■ Ensemble avg 0.013

■ GAM avg 0.323

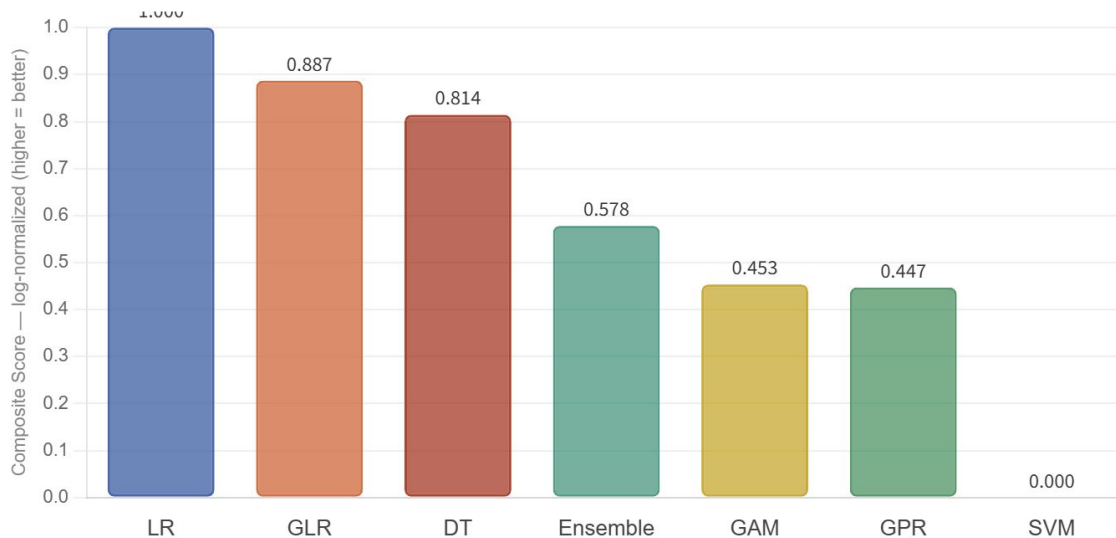
■ Mean



Model	DO_1	DO_2	DO_3	NO3	NH4	Avg
LR	0.203	1.000	0.990	0.999	1.000	0.838
GLR	0.394	0.547	1.000	1.000	0.775	0.743
GPR	0.587	0.258	0.150	0.776	0.436	0.441
GAM	1.000	0.001	0.136	0.213	0.267	0.323
DT	0.000	0.000	0.018	0.090	0.043	0.030
Ensemble	0.000	0.066	0.000	0.000	0.000	0.013

Among the five parameters, LR has the highest accuracy for four of them, making it the most accurate model overall. However, if precise prediction of DO_1 is needed, the data provided by GAM should be referred to; for the rest, LR should be the reference.

Overall, LR ranks first in both efficiency and accuracy, making it the best-performing model. DLR is slightly lower in both metrics, placing it second. The DT model sacrifices accuracy for higher efficiency, so it is a worse model. Among the remaining four models, aside from the very poor SVM, the other three show only average performance.



Rank	Model	Avg RMSE	Avg Time (s)	RMSE × Time	Score
1	LR	1.0913	0.0053	0.0058	1.0000
2	GLR	1.1032	0.0181	0.0199	0.8872
3	DT	1.2997	0.0341	0.0444	0.8144
4	Ensemble	0.7704	0.7704	0.5935	0.5782
5	GAM	1.2298	1.8991	2.3356	0.4534
6	GPR	1.1574	2.1639	2.5044	0.4471
7	SVM	14.0703	24.1321	339.5455	0.0000

* Log transformation compresses the extreme gap caused by SVM while preserving rank order and visually surfacing differences among the remaining models.

Conclusion

Hypothesis: If the appropriate model is used and the data is properly standardized, then the RMSE will be reduced. A better model will allow for better outcome predictions for the wastewater treatment plant.

Conclusion: A wastewater treatment plant should use LR with a GAM supplement for their prediction model, but normalization doesn't matter. One model is too simple and not enough to be optimal for every parameter.

Future Work: Testing other algorithms with more data



References

Afreen Nishat, Yusuf, M., Qadir, A., Yassine Ezaier, Vambol, V., M. Ijaz Khan, Sana Ben Moussa, Hesam Kamyab, Sehgal, S. S., Prakash, C., Yang, H.-H., Ibrahim, H., & Eldin, S. M. (2023). Wastewater treatment: A short assessment on available techniques. *Alexandria Engineering Journal*, 76, 505–516.

<https://doi.org/10.1016/j.aej.2023.06.054>

Alvi, M., Batstone, D., Mbamba, C. K., Keymer, P., French, T., Ward, A., Dwyer, J., & Cardell-Oliver, R. (2023). Deep learning in wastewater treatment: a critical review. *Water Research*, 245, 120518.

<https://doi.org/10.1016/j.watres.2023.120518>

Crini, G., & Lichtfouse, E. (2018). Advantages and disadvantages of techniques used for wastewater treatment. *Environmental Chemistry Letters*, 17. <https://doi.org/10.1007/s10311-018-0785-9>

